

# Elsa Elisa Yohana Sianturi

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Informatics Engineering graduate focused on AI Engineering, Natural Language Processing, Computer Vision, and Multimodal Learning. Experienced in building RAG-based document question-answering systems, transformer-based NLP models, and CLIP-based multimodal classifiers. Skilled in PyTorch, Hugging Face, Scikit-learn, OpenCV, and LLM integration, with hands-on experience deploying AI prototypes using Gradio and Hugging Face Spaces

## EDUCATION

**Bachelor's Degree in Informatics Engineering | Institut Teknologi Sumatera**  
Lampung, Indonesia

Aug 2022 - May 2026

- GPA: 3.64/4.00
- Focus: Deep Learning, Machine Learning, NLP, and Computer Vision
- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Multimodal Machine Learning, Data Visualization and Information

## EXPERIENCE

**AI Engineer Intern — PT Pertamina Hulu Rokan**  
Pekanbaru, Indonesia

Jun 2025 - Jul 2025

- Developed AIDocs, a RAG-based question-answering prototype that accelerates information retrieval from PDF land agreements, eliminating the need for manual document checks
- Engineered the end-to-end data pipeline, covering document preprocessing, hybrid retrieval, answer generation, and source citation utilizing LangChain, FAISS, and BM25
- Evaluated and optimized system performance against original document content, achieving 73.33% answer accuracy and successfully delivering an AI-driven solution for document digitalization

**Teaching Assistant — UPA TIK ITERA**  
Lampung, Indonesia

Aug 2023 - Present

- **Operating Systems**  
Facilitated hands-on laboratory sessions focusing on process management, memory allocation, and operating system fundamentals
- **Introduction to Computing & Software I (PKS I)**  
Guided students in procedural programming, computational thinking, and fundamental programming logic  
*Credential ID: B/5359/IT9.E2/TA.00.00/2024*
- **Interaction Design**  
Provided technical mentoring for student UI/UX projects, user-centered design principles, and project-based assignments  
*Credential ID: 33133/IT9.3.3/TA.00.00/2025*

## PROJECTS

**Multimodal Deep Learning for Self-Harm Meme Detection | [GitHub]**  
Tech: CLIP, ELECTRA, Gradio

Aug 2025 - Apr 2026

- Developed a multimodal binary classification system for a final thesis project to accurately identify self-harm versus non-self-harm memes using combined visual and textual representations
- Engineered an intermediate fusion architecture via concatenation, leveraging CLIP ViT-B/32 for image encoding and ELECTRA-base for text encoding
- Trained and evaluated the model on a stratified dataset of 2,024 memes, achieving an outstanding 96.1% F1-score and successfully outperforming both single-modality baselines (CLIP at 91.5% and ELECTRA at 92.0%)
- Deployed an interactive web-based inference demonstration of the model utilizing Gradio on Hugging Face Spaces

**Multimodal City Branding Analysis | [GitHub]**  
Tech: Image-Text Representation Learning, Social Media Analysis

May 2026 - Present

- Developed a multimodal deep learning model to analyze South Lampung tourism image and caption data from Instagram

- Used openai/clip-vit-base-patch32 for image representation and cardiffnlp/twitter-xlm-roberta-base for text representation
- Achieved approximately 89.2% validation accuracy at the selected best epoch based on validation loss
- Collaborated with researchers and students from Urban and Regional Planning and Informatics Engineering

#### Student Face Recognition with ArcFace and SVM | [GitHub]

Jun 2026

Tech: InsightFace ArcFace, Scikit-learn, OpenCV, SVM, Jupyter Notebook

- Developed an end-to-end facial identification system to automate the recognition of 70 student identities from uploaded facial images
- Engineered a pipeline using offline image augmentation, 512-dimensional InsightFace ArcFace Buffalo\_L embeddings, and an RBF-kernel SVM classifier with leakage-resistant Stratified Group K-Fold evaluation
- Processed 1,946 facial samples and achieved 82.0% accuracy, 85.4% weighted precision, and 82.4% weighted F1-score

## LEADERSHIP & ORGANIZATIONS

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### Head of Academic & Scholarship Division — HMIF ITERA

Jan 2025 – Dec 2025

Lampung, Indonesia

- Led 4 academic and scholarship-related programs, served as a speaker in a scholarship-sharing session, and managed mentoring initiatives that contributed to 2 students receiving the Pertamina Sobat Bumi Scholarship

### Secretary of Minister of Information Technology — KM ITERA

Apr 2024 – Dec 2024

Lampung, Indonesia

- Managed meeting notes, project documentation, and GitHub-based timeline tracking to coordinate IT-related student government web projects and report progress to the Minister of Information Technology

### Head of Entrepreneurship Department — Sobat Bumi Lampung

Oct 2025 – Present

Lampung, Indonesia

- Coordinated vendor communication and production planning for Sobat Bumi Lampung merchandise, including jackets, vests, and community attributes, while supporting entrepreneurship initiatives within the scholarship community

## SKILLS

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- **Programming:** Python, SQL
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, Hugging Face, OpenCV, Pandas, NumPy, Matplotlib
- **NLP & LLM:** LangChain, FAISS, BM25, ChromaDB, OpenAI API, RAG, Transformer-based Models
- **Computer Vision:** CLIP, EfficientNet-B8, ResNet-50, Image Classification, Multimodal Learning
- **Model Development:** Fine-tuning, Hyperparameter Tuning, Ablation Study, Error Analysis, Model Evaluation
- **Deployment & Tools:** Gradio, RunPod, Hugging Face Spaces, FastAPI, Git/GitHub, Linux, W&B, Figma, Tableau

## CERTIFICATIONS

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- **AWS re/Start Cloud Computing Program** - Orbit Future Academy & Amazon Web Services (Mar 2026 - Present)
- **Belajar Pemrograman Prosedural dengan Python** - Dicoding (2024) | Credential ID: 2VX3O4W23ZYQ
- **Memulai Pemrograman dengan Python** - Dicoding (2024) | Credential ID: L4PQQ8O2PO01
- **Belajar Dasar Structured Query Language (SQL)** - Dicoding (2024) | Credential ID: NVP77M3WVPR0
- **AI Praktis untuk Productivity** - Dicoding (2024) | Credential ID: 1OP82QOYLPQK

## AWARDS & HONORS

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- **Pertamina Sobat Bumi Scholarship Recipient** - Pertamina Foundation (2024)
- **2nd Winner of UI/UX Competition Point Project 2.0** - HMIF ITERA (2024) | Credential ID: 313/525/SRT/HMIF/XII/2024

## LANGUAGES & PUBLICATIONS

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- **Languages:** Indonesian (Native), English (Professional Working Proficiency)
- **Publications:** Sianturi, E. E. Y., et al. "Transformasi Digital Layanan Posyandu." PADIMAS, 2025 | [Full Paper]